UNIVERSITY OF AUCKLAND
Waipapa Taumata Rau
NEW ZEALAND

SCIENCE

COMPSCI 714
Lecture 1: Welcome to the course!

Course structure and a short approach to AI

Instructors: Thomas Lacombe & Qian Liu

What to expect?

- In this course, you will:
  - Gain knowledge about underliying methods used to develop AI systems.
  - Get practical experience in training and evaluating AI systems based on Deep Neural Networks (DNNs), using well known frameworks.
  - Get familiar with well known DNNs architectures used for different type of data.
  - Gain knowledge and get practical experience with some of the latest developments in the field (Generative AI, Multimodal AI).
  - Get insights from industry experts with several guest lectures and tutorials.

Teaching team

Qian Liu (course coordinator)

Lecturer for the second half of the course

Room: 903-422 (Newmarket Campus)

Office hours: On request

liu.qian@auckland.ac.nz

Thomas Lacombe (course director)

Lecturer for the first half of the course

Room: 303-419 (City Campus)

Office hours: On request

thomas.lacombe@auckland.ac.nz

Xiao Li

Course Tutor/Marker

xli990@aucklanduni.ac.nz

Ning Wang

Course Tutor/Marker

ning.wang@auckland.ac.nz

Junyi Chen

Course Tutor/Marker

junyi.chen@auckland.ac.nz

Thomas' background

UNIVERSITÉ SAVOIE MONT BLANC

Communauté UNIVERSITÉ Grenoble Alpes

PhD in E.Eng. in 2018, at USMB/CUGA, France

- Industrial computer vision and ML for automatic visual quality control
- Feature engineering, feature selection

Postdoc at UoA in 2019-2020

- Automated hyper-parameter tuning for machine learning with data streams

https://scikit-ika.github.io/

Since 2021

- Teaching fellow in the School of Computer Science
- Interest in interactions between Generative AI and education

Qian Liu

Lecturer for the second half of the course

Room: 903-422 (Newmarket Campus)

Office hours: On request

Email: liu.qian@auckland.ac.nz

Academic Qualifications

- PhD in Computer Science (2020) at Beijing Institute of Technology, China
- PhD in Software Engineering (2022) at University of Technology Sydney, Australia

Work Experience

- 2021-2024, Postdoc Research Fellow at Nanyang Technological University, Singapore.
- Since 2024, Lecturer in School of Computer Science at University of Auckland, New Zealand

Research Topics

- Natural Language Processing
- Reliable Large Language Models

北京理工大學 BEIJING INSTITUTE OF TECHNOLOGY

UTS UNIVERSITY OF TECHNOLOGY SYDNEY

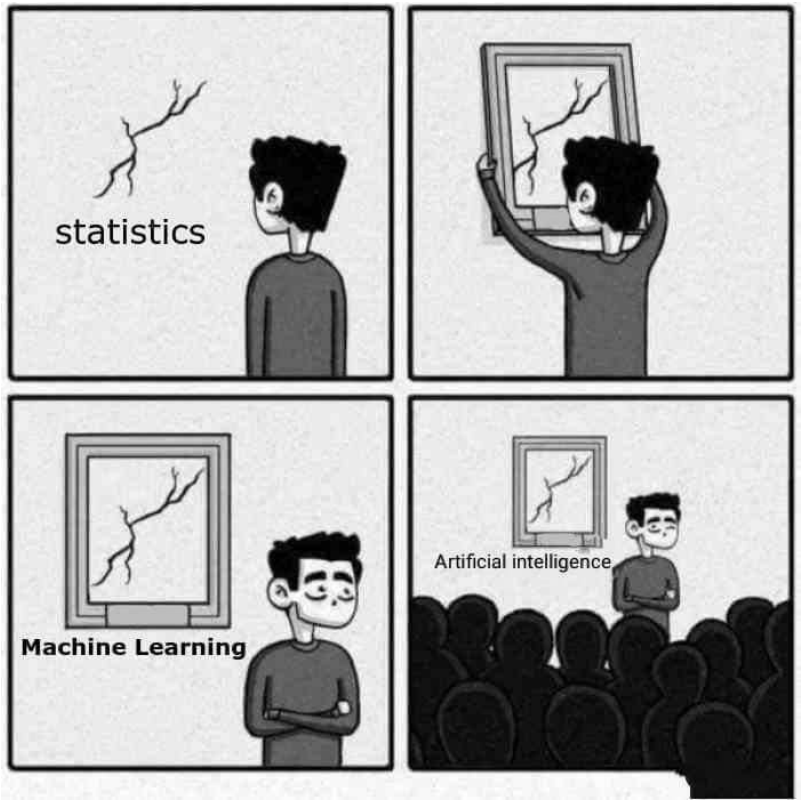
NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

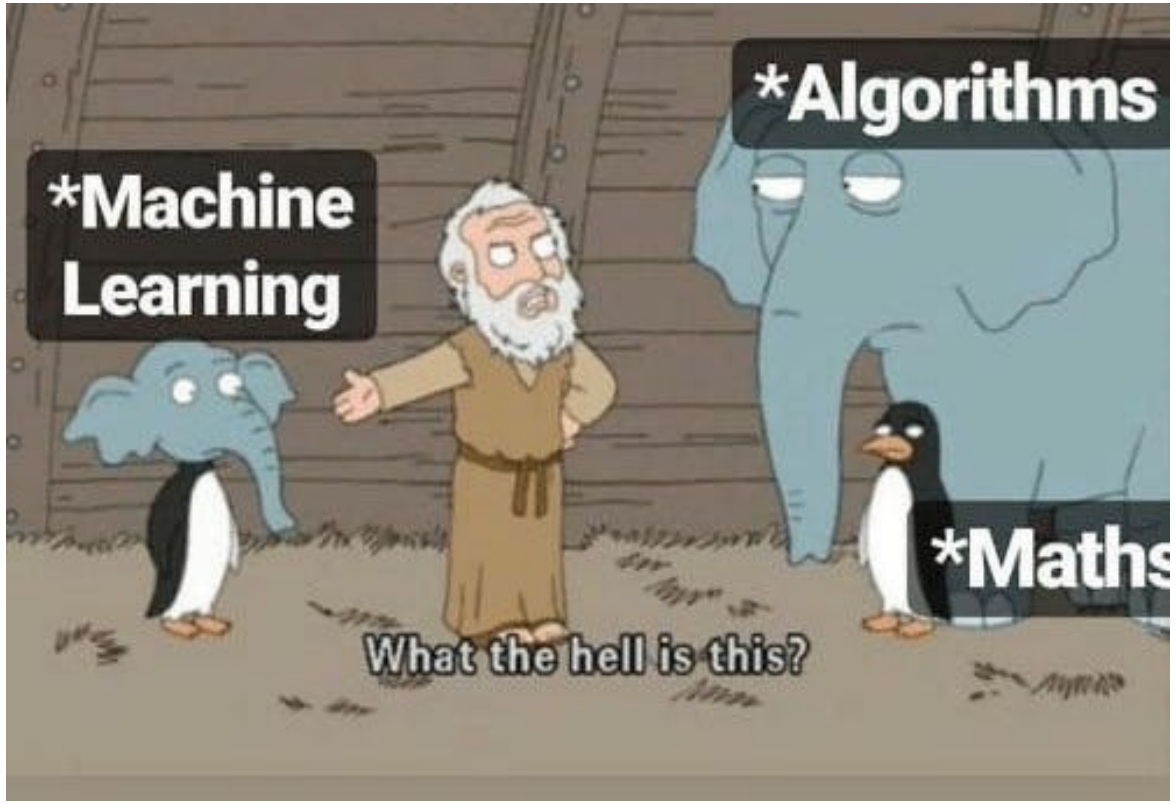
Waipapa Taumata Rau University of Auckland

Before we start: course pre-requisites

- We designed this course to be hands-on and practice oriented.
- We expect students to be familiar with Python and coding using notebooks (locally on your machine or on Google Colab).
- Make sure you follow the instructions in the Canvas Hei tīmata | Start Here page to get set up!
- If you have limited experience with Python, we highly recommend you go through some of the tutorials listed in Canvas.
- Some base knowledge of math concepts covering calculus, linear algebra, probabilities, and statistics, is also recommended.

AI: a math story?





Join at menti.com | use code 7233 7209

Mentimeter

How would you evaluate your programming skills before starting COMPSCI 714?

Programming skills in at least one programming language.

Programming skills in Python.

Programming skills in Python for data science and/or machine learning.

Novice (you are new to this)

Very experienced

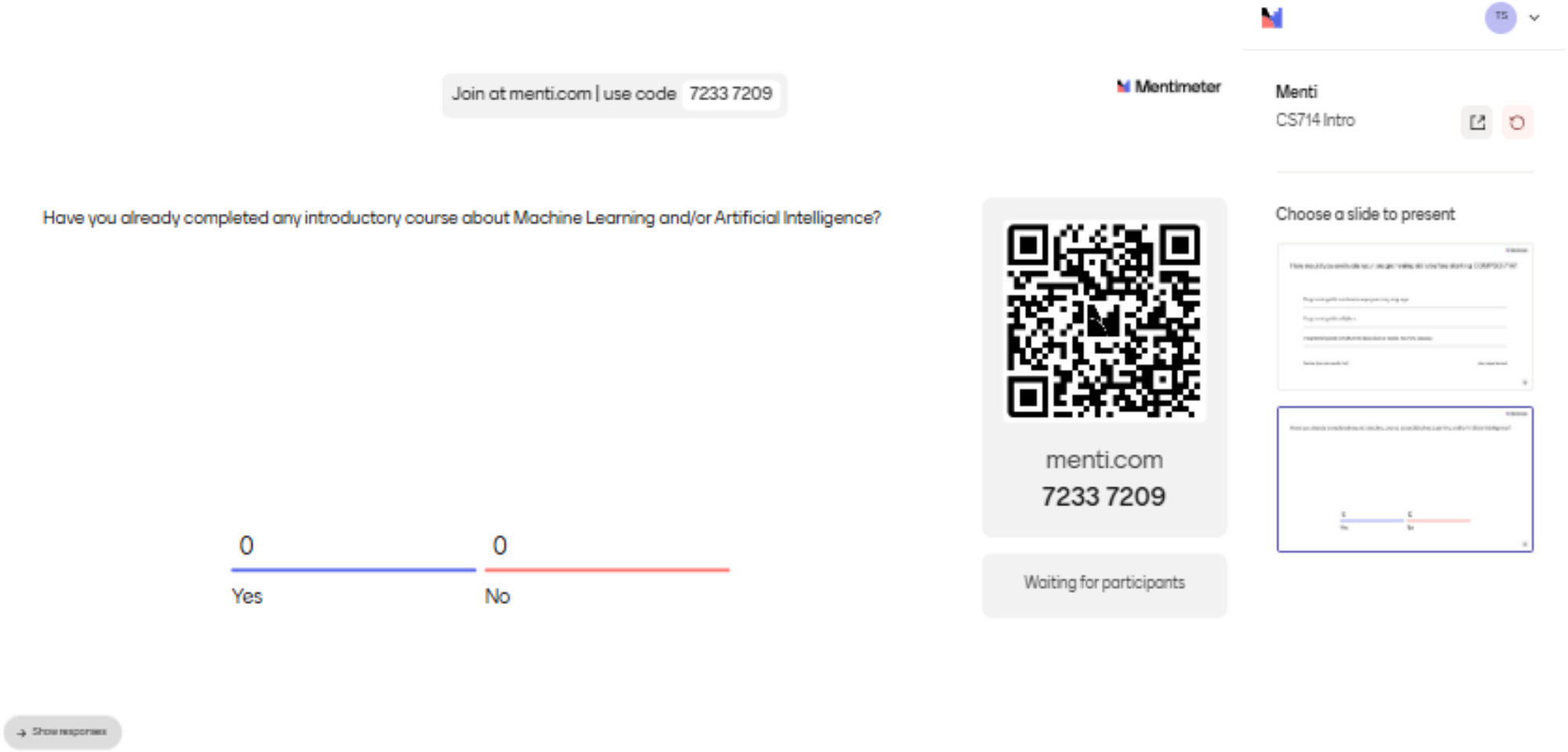
Show responses

TS

Menti

CS714 Intro

Choose a slide to present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Mentimeter
Menti
CS714 Intro
Choose a slide to present
Have you already completed any introductory course about Machine Learning and/or Artificial Intelligence?
menti.com
7233 7209
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Yes
No
Waiting for participants

Admin I: Communication in the course

- **Teaching team to students**:

  Canvas Announcements
  - Check them/your UoA email daily (please make sure that your personal notification settings is set to ASAP for announcements).

- **Students to teaching team**:

  Ed Discussion Forum & Email
  - Ed Discussion forum for general course questions, e.g., content related, general admin question.
  - Email for specific/private/urgent requests/meeting request.

- **Email guidance**:
  - Email the right person: course coordinator (admin-related) vs. lecturer (content-related)
  - Use descriptive email subjects: [CS714] subject of the email
  - Be short & to the point, while providing enough background information.
  - Make sure the answer to your question is not given on the Canvas home page, class announcements, or the class discussions on Piazza before emailing.

- Please follow the school guidelines for communication and course expectations.

Admin II: Lecture/tutorials

When	Where
Wednesday 3pm to 4pm (lecture)	405-460
Thursday 4pm to 6pm (lectutorial)	BLT100 (106-100)

- In general:
  - Wednesday: Lectures (in-person)
  - Thursday: Lectutorial (in-person)
- This might vary for some weeks (see up-to-date info in the Course Schedule).

- Bring a personal laptop in lectutorials (at least) as there will be programming activities! Notebooks will be shared on Canvas.

Admin III: Assessments

Two Lane Assessment

- **Lane 1** consists of controlled assessments *of* learning. These tasks measure what you are expected to know **without** the assistance of GenAI after completing the course.

- **Lane 2** consists of uncontrolled assessments *for* learning. These tasks are designed to support your learning as you complete them. You are allowed to use GenAI, provided that your use is **responsible, ethical, and transparent**.

- More in two-lane assessment framework

Admin III: Assessments

Assessment Group	Weighting	Assessment Lane
Coding Assignments	30% (2 x 15%)	Lane 2 (assessment for learning)
Group Project (hackathon-based)	30%	Lane 2 (assessment for learning)
Midsemester Test	15%	Lane 1 (assessment of learning)
Final Exam	25%	Lane 1 (assessment of learning)

- To pass the course, you need:
  - 50% overall
  - 50% of the Lane 1 group (coding assignments + group project), i.e. 30/60 over the 2 assignments + the group project.
  - 50% of the theory part (mid-semester test + exam), i.e. 20/40 over the test + the exam.
- Assignments and group project instructions will be released along the semester.

Coding Assignments

- Designed to help you better understand concept covered in the lectures and evaluate your individual coding skills applying or adapting code covered in the lectutorials, but also prompting you to go beyond course content.

- GenAI policy:
  - You may use Generative AI tools to support your learning while working on coding assignments. Acceptable uses include clarifying concepts or code from the lectutorial notebooks, assisting with debugging, explaining error messages, exploring alternative implementation approaches, or helping you better understand relevant documentation.
  - However, Generative AI **must not be used to directly generate or substantially complete solutions to the assessed tasks**. You remain fully responsible for the design, implementation, testing, and correctness of your own code. Any use of Generative AI should **support your understanding rather than replace your problem-solving process**.

Hackathon-based group project

WHY (this project)?
Designed to give you hands-on experience in designing an AI solution to a real-world problem, using professional tools and mindset.

WHO (is involved)?
- COMPSCI 714 students and teaching team
- University of Auckland Research and Innovation Office (RIO)
- UoA staff (use case owners)
- Centre for Innovation and Entrepreneurship (CIE)
- Microsoft

Hackathon-based group project

WHAT (to expect)?

1. **Form a team** with complementary skills and assign roles
2. **Attend some training** (some in person, some online) to develop soft skills (e.g., design thinking, teamwork, prototyping) and technical skills (Microsoft Azure/Foundry tools)
3. **Attend webinars from use case owners** and choose a use case to focus on
4. Understand the use case and **design a relevant solution**
5. **Implement a proof of concept**/prototype to showcase your solution using constrained resources (tools, time, budget, data)
6. **Pitch your solution** and **write a report** outlining its design and justification

Hackathon-based group project

WHEN (things happen)? Tentative project schedule

Week	Event/Activity
Week 1 (starting 02/03)	Introduction of the project structure and expected deliverables in the course intro lecture
Week 2 (starting 09/03)	Introduction of the project requirements with high level use case descriptions (TBC)
Week 3 (starting 16/03)	• **CIE in-person Workshop**: Solution design / Prototyping - **Monday 16/03 2pm to 4pm** • **Microsoft online Workshop**: Tools available for the project and how to get started (TBC)
Week 4 (starting 23/03)	Webinars with use case owners/experts
Week 5 (starting 30/03)	Webinars with use case owners/experts
Mid-semester break	Teams work on ideation
Mid-semester break	Teams work on ideation
Week 6 (starting 20/04)	Teams work on their solution design
Week 7 (starting 27/04)	Teams work on their solution design
Week 8 (starting 04/05)	Teams work on their solution design
Week 9 (starting 11/05)	Teams work on implementing their Proof of Concept
Week 10 (starting 18/05)	Teams work on implementing their Proof of Concept
Week 11 (starting 25/05)	Teams work on implementing their Proof of Concept
Week 12 (starting 1/06)	• **Final event**: Group presentations with jury (tentative, Saturday 6/06) • Reports due end of week 12

Hackathon-based group project

Industry-focused AI capabilities

- Analysing an ill-defined problem to identify key requirements and constraints
- Designing a relevant, client-centred AI-based solution
- Implementing a proof of concept using appropriate tools within practical constraints (e.g., budget, time, technical limitations, data privacy)
- Managing an AI project, including planning, coordination, and iterative development
- Presenting your solution and proof of concept in a professional, clear, and persuasive manner

Hackathon-based group project

Non-graded deliverables (all descriptions will be published soon)

- **Team registration** with primary roles assignment – 22/03
  - Teams of **5** students with following roles:
    1. Project lead
    2. Data & Problem Analyst
    3. Solution Architect
    4. Implementation Lead
    5. AI Governance Lead

- **Use case choice registration** – 5/04 (use cases descriptions released next week + use case owners webinars in weeks 4 & 5)
- **Project checkpoint 1** – 26/04 (content TBA)
- **Project checkpoint 2** – 17/05 (content TBA)

Hackathon-based group project

Graded deliverables (rubrics and templates will be provided soon)

- **Group presentation** (Saturday 6/06, tentative) (marked by the jury)
  - Pitch your solution, including a demo of the proof of concept to a jury composed of member of:
    - The COMPSCI 714 teaching team,
    - The Research and Innovation Office,
    - Microsoft.

- **Group report** (due Sunday 7/06) (marked by the COMPSCI 714 markers)
  - Explain acceptance and success criteria you defined after interacting with stakeholders and assumptions you made
  - Describe and justify your solution architecture more in depth
  - Provide your project timeline and explain team organisation
  - Detail the next steps and expected costs to a full solution

Hackathon-based group project

Save the dates

- **CIE in-person Workshop**: Solution design / Prototyping - **Monday 16/03 2pm to 4pm** (Unleash space, Faculty of Engineering)
  - Great opportunity to meet classmates and form teams!

- **Final event**: Group presentations with jury (tentative, **Saturday 6/06**, Unleash Space)
  - Each team will be attributed a presentation time closer to the date.
  - Presentations will be in person.
  - Come for your presentation, but you are encouraged to stay to watch other teams as well.

Mid-semester test

COMPSCI 714 Mid-semester Test (15%)

- **Start Time and Date**: Tuesday 21th of April at 6:30pm
- **Duration**: 60 min (55min + 5min reading time)
- **Where**: On city campus. Rooms TBA.
- **Type of Test:** On paper.
- **Content covered**: All lectures up to the mid-term break (weeks 1 to 5 included).
- **What you must take with you to the test room:** A pen and your Student Id card.
- **What else can you can take with you to the test room:** Bottled water, and an A4 2-sided sheet of handwritten notes.
- Calculators are not allowed.

Final exam

COMPSCI 714 Final Exam (25%)

- **Start Time and Date**: TBA during the semester (will take place in the exam period after week 13)
- **Duration**: 2 hours
- **Where**: On campus. You will be able find your room on SSO.
- **Type of Test:** On paper.
- **Content covered**: All lectures, except industry guest lectures
- **What you must take with you to the exam room:** A blue or black pen (plus a spare), a pencil, and your **Student Campus Card (a $25 fine applies if you don't have it with you)**.
- **What else can you can take with you to the test room:** Bottled water, and **2 double-sided A4 pages of handwritten notes** (your mid-semester one and a new one). The page can be printed if you wrote your notes on a tablet, but typed noted will not be accepted (same reasons as mid-semester test).
- Calculators are not allowed.

Thomas
Qian
Guest
No lecture/tutorial

Admin IV: Tentative schedule and topics

Week	Wednesday (3pm - 4pm) – Room: 405-460	Thursday (4pm - 6pm) - Room: BLT100 (106-100)
1	Intro lecture (all the team is welcome to join)	Lectutorial: Data acquisition / loading / pre-preprocessing
2	Lecture: NN and DNN basics	Lectutorial: Training and evaluating a DNN
3	Lecture: Design choices	Lectutorial: Hyperparameter tuning and regularisation
4	Lecture: Architectures for Sequential data	Lectutorial: RNNs and Transformers
5	Lecture: Architectures for Image data	Lectutorial: Convolutions, CNNs & ViT
Break	2 weeks	
6	Guest Lecture: ZEIL (TBC)	Guest lectutorial: Snowflake (TBC, **date might change**)
7	Lecture: Transfer Learning	Lectutorial: Transfer Learning
8	Lecture: Transfer Learning (Pretrained Foundational Models)	Lectutorial: Transfer Learning and Finetuning Pretrained Models
9	Lecture: Generative AI	Lectutorial: Generative AI
10	Lecture: Generative AI and Large Language Models	Lectutorial: Generative AI and Large Language Models
11	Lecture: Multimodal AI	Lectutorial: Multimodal AI
12	Guest lecture: ASB	Guest lecture: TBA

Class rep

- The class rep act as an intermediary between students in the class and the lecturers and tutor.
- You can share with them any suggestions / complaints / remarks about the lectures.
- Class rep can organise communication between students (e.g. create social media chats).

- It is mandatory to have one if the course has at least 15 students.
- The role of a lecturer is to facilitate the class rep selection process and not to impose a candidate to the class.

- We will run the class rep election on Ed Discussion!
- I will open a post where candidates can post their motivations. Then, we will proceed to a vote with a poll on Canvas if there are several candidates.

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Do you need help?

The University has a range of ways to get help if something has happened to you or someone you know.

Te Papa Manaaki, Campus Care

A safe, confidential and free service where a case manager will connect and guide you through relevant support services. You can be referred by others or reach out directly for advice.

auckland.ac.nz/campus-care

University Health and Counselling Services (UHCS)

Enrolled patients can book an appointment with a GP, nurse or counsellor. Visit the website for more info, including online enrolment.

auckland.ac.nz/studenthealth

Ratonga Hauātanga Tauira, Student Disability Services (SDS)

The team offer a range of services to make studying at the University an accessible and positive experience by tailoring support to suit each individual student.

disability.auckland.ac.nz.

Be Well

A range of general wellbeing self-help resources, to support your physical and mental health during your time at University.

auckland.ac.nz/be-well

Student Conduct Office

The primary reference point for all matters relating to student general misconduct, including your options for reporting and available support.

auckland.ac.nz/student-conduct

Complaints and incidents

This is where you can submit a complaint, provide feedback or report an incident to the University.

auckland.ac.nz/complaints-and-incidents

Alcohol and Drug Clinician

Confidential alcohol and drug advice and appointments are available for all students.

auckland.ac.nz/aod-clinician

Rainbow Support Adviser

Provides a holistic approach to student wellbeing for the rainbow community, offering 1:1 support and group opportunities for discussion.

equity.auckland.ac.nz/rainbow

Spiritual and religious support

The team can provide holistic pastoral care and wellbeing. They offer inclusive, compassionate support whether you have a faith or are exploring how spirituality and religion could be a part of your life.

auckland.ac.nz/spiritual-support

AUSA Advocacy

The team provide free, independent, professional advice if you are struggling with an academic, financial or personal issue.

ausa.org.nz/support/advocacy

HELP Centre

A private and confidential service providing specialist support for sexual abuse survivors. Search 'HELP' on the University website for information on the drop in service.

Phone: 0800 623 1700 (available 24/7)
helpauckland.org.nz

Puāwaitanga Counselling

A service run by trained counsellors and available to all students, providing support with feelings of anxiety, relationships, grief, addiction and more.

Phone: 0800 782 999 (7 days a week 9am-9pm)
Visit auckland.ac.nz and search 'Puāwaitanga'.

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Be Well
Whai Hauora

Your To-do list

- If not done already, get familiar with the course Canvas page, and join the Ed Discussion forum for the course.
- Make sure you are up-to-date with the programming prerequisites before Tutorial 1 on Thursday.
- Make sure you have a laptop and you installed the necessary software (see on Canvas).
- For potential class reps: submit your motivations on Ed Discussion (once I create the post).

Course and extra resources

- Books / Websites / Repositories (see Course Resources & Recommendations Canvas page)
  - First half of the course leverages content from:

    Géron, A. (2025). *Hands-on machine learning with Scikit-Learn and PyTorch*. O'ReillyLinks (Full text available online via UoA library)

- The Batch - Newsletter (DeepLearning.ai) → Weekly AI quiz (not graded) in Wednesday lecture

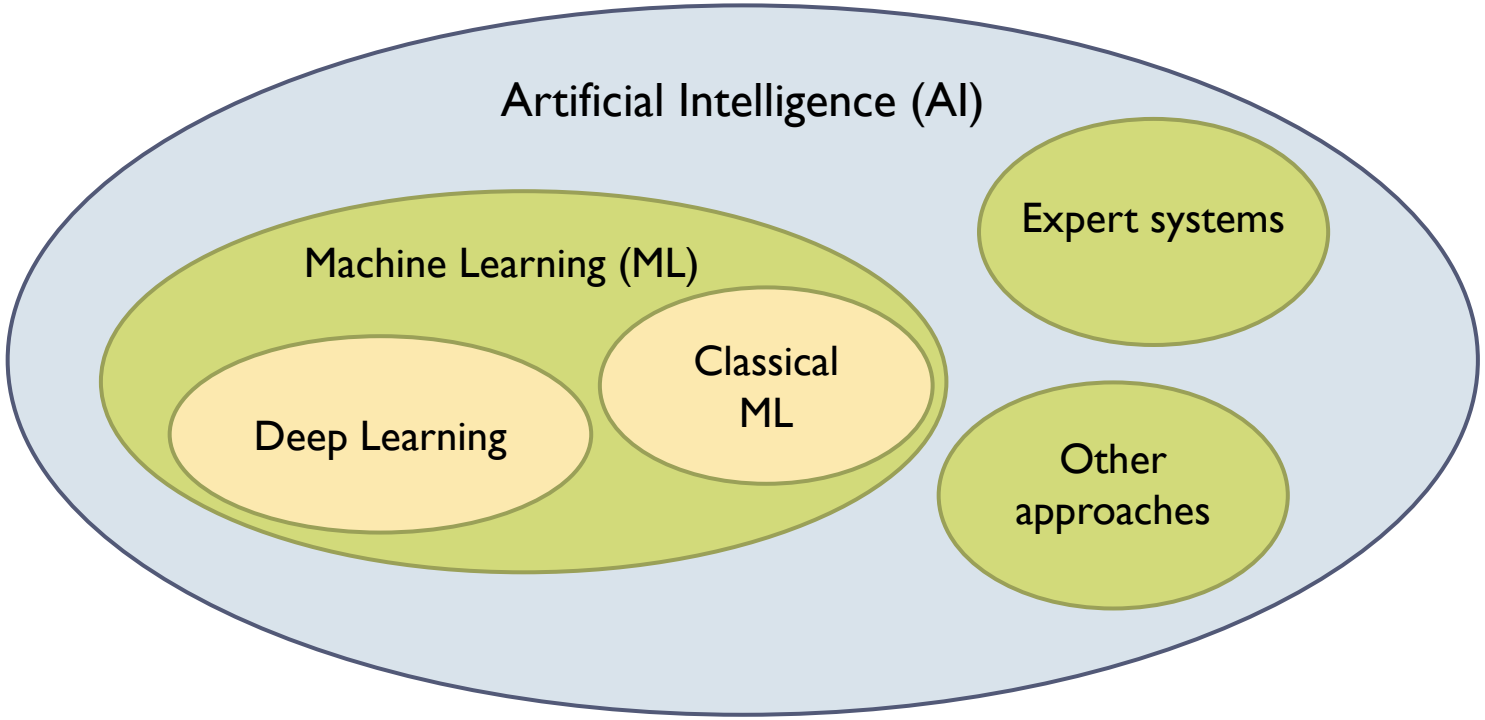
THE BATCH
What Matters in AI Right Now

Your feedback is valuable!

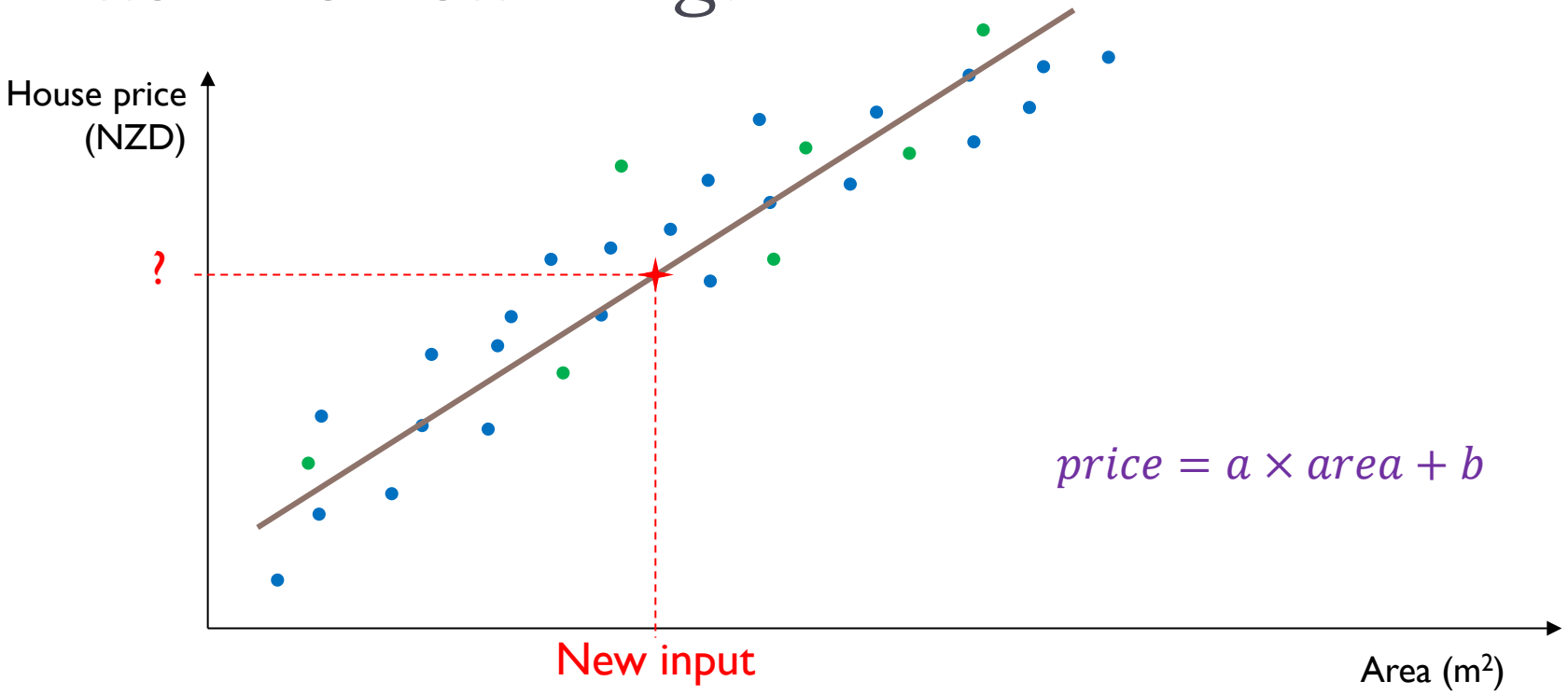
- This is the second iteration of the course, so we welcome your feedback to improve it in the future ☺
- Feedback should preferably be passed by:
  - Talking to your class rep so that they can report it to us
  - Filling the SET Evaluations (at the end of the semester)

Let's define AI first

- Artificial Intelligence: "The science and engineering of making intelligent machines", John McCarthy, 1955
- Machine Learning: Field of study focusing on algorithms which enable the computers to learn from data, and even improve themselves, without being explicitly programmed.
- Deep Learning: subset of machine learning methods based on artificial neural networks (ANNs) with representation learning.

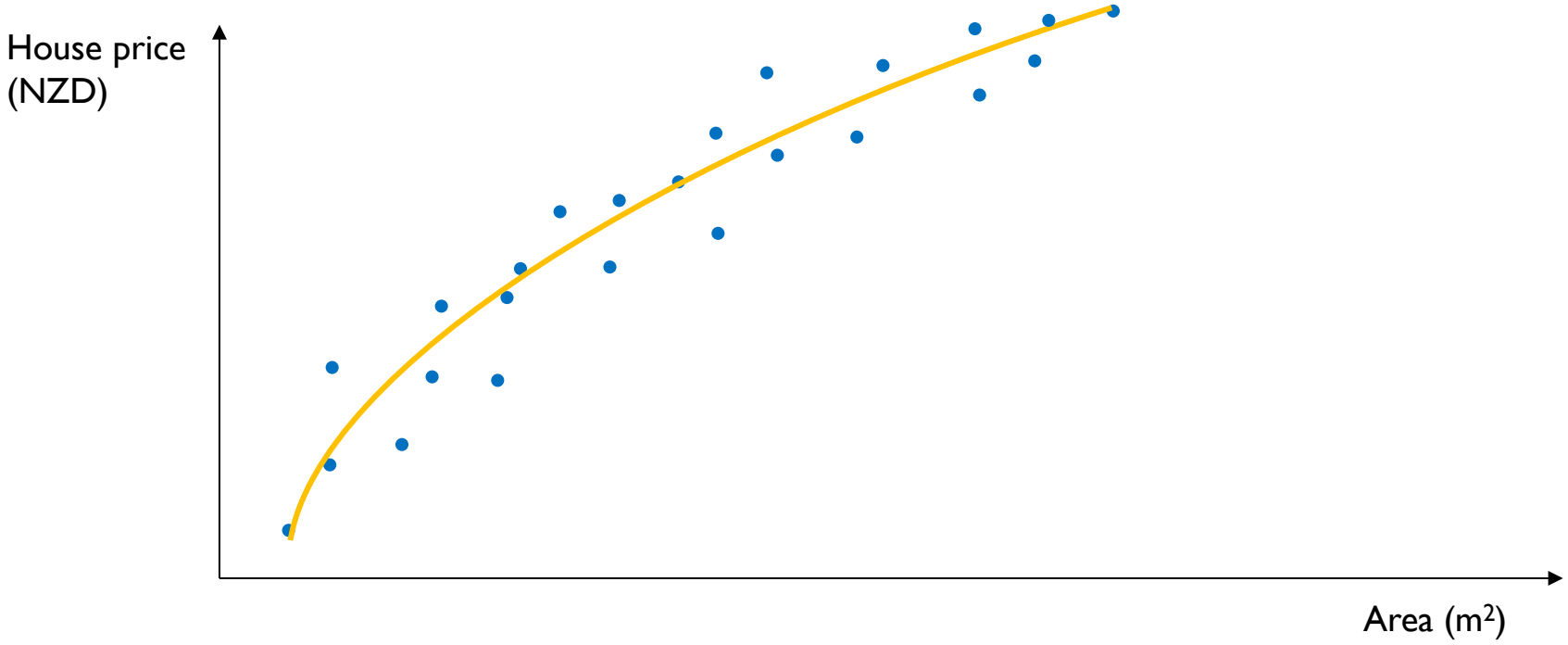


What is Machine Learning?



$$price = a \times area + b$$

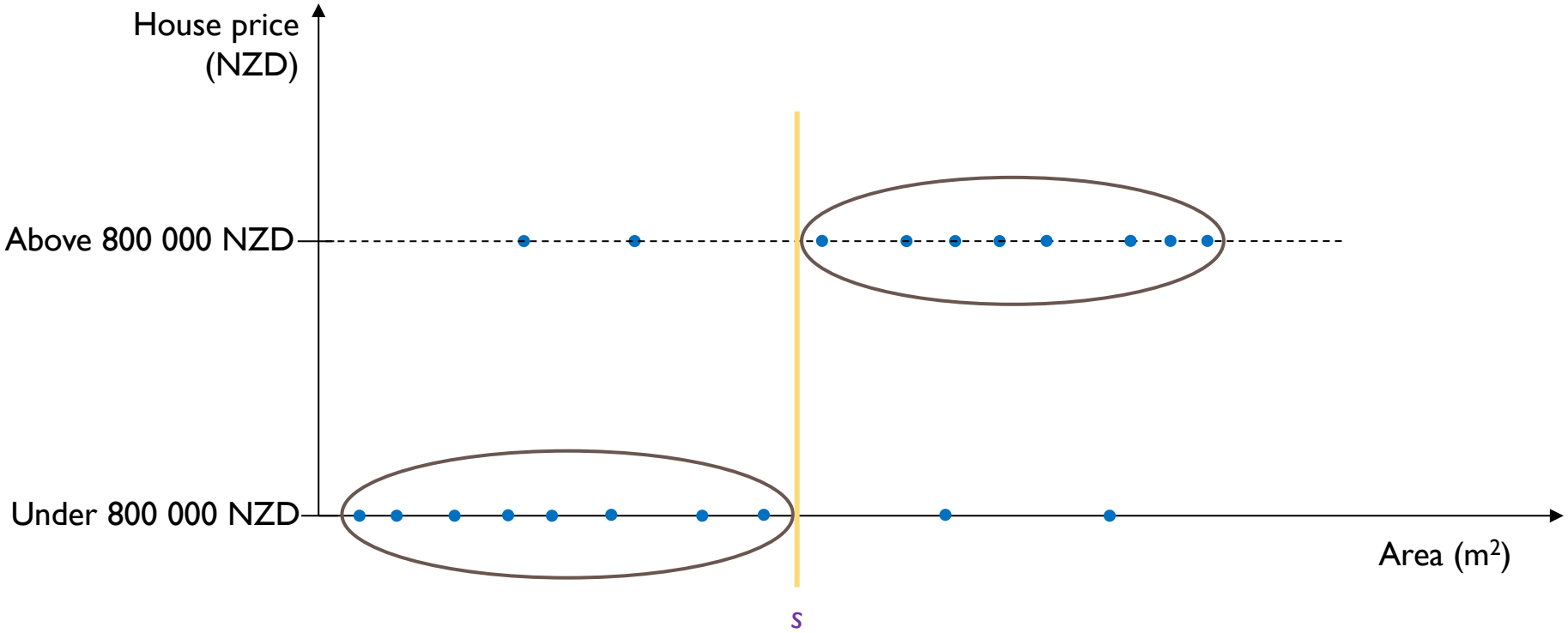
1) ML algorithms aim at uncovering *patterns* in a set of *training data* (e.g., mapping inputs and outputs).
2) A *model* is created out of the uncovered patterns.
3) The model's performance is tested with *test data*.
4) If performances are satisfactory, the model can be used to predict new inputs.



- Relationship between inputs and outputs can be linear or not.
- Several types of ML algorithms can be used to build models (e.g., Linear Regression).

Classification vs Regression problems

- Regression problem → predicting continuous/numerical values.
- Classification problem → predicting discrete/categorical values.



Types of Machine Learning

Supervised learning

- Train a model for a specific task using labelled data (e.g., classify cats and dogs images).
- Can be used for classification or regression.
- Methods include: Linear Regression, K-Nearest Neighbor, Decision Tree, Random Forest, Support Vector Machine, Neural Network.

Unsupervised learning

- Train a model to discover patterns and relationships in unlabelled data (e.g., identify different customers behaviours).
- Clustering methods include: k-means, DBSCAN.
- Dimensionality reduction methods include: PCA, LDA, t-SNE.

Reinforcement learning

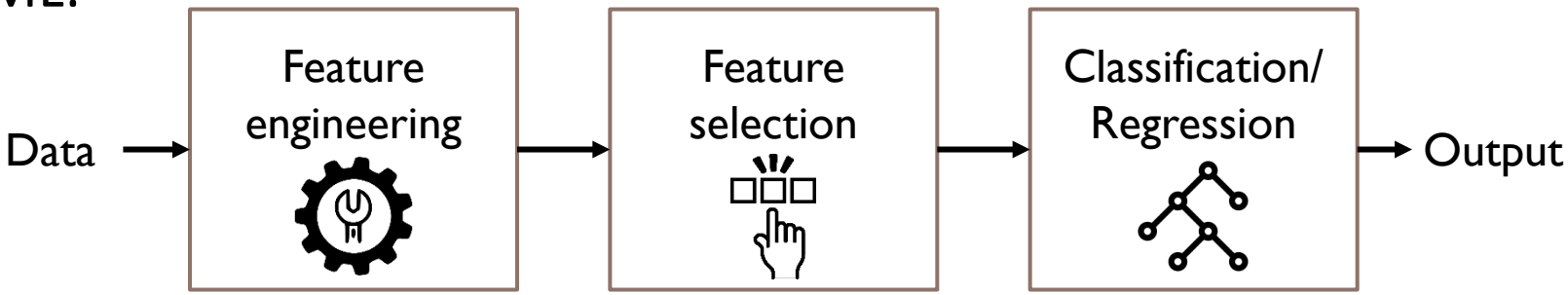
- Let an agent perform actions in an environment to maximize a reward (e.g., training a robot to move from point A to point B).
- Methods include: Monte Carlo methods, Q-learning.

Self-supervised learning

- Similar to supervised learning, but the labels are self-generated instead of given by humans.
- E.g., masking words sentences and training to predict the masked words.

Classical Machine Learning

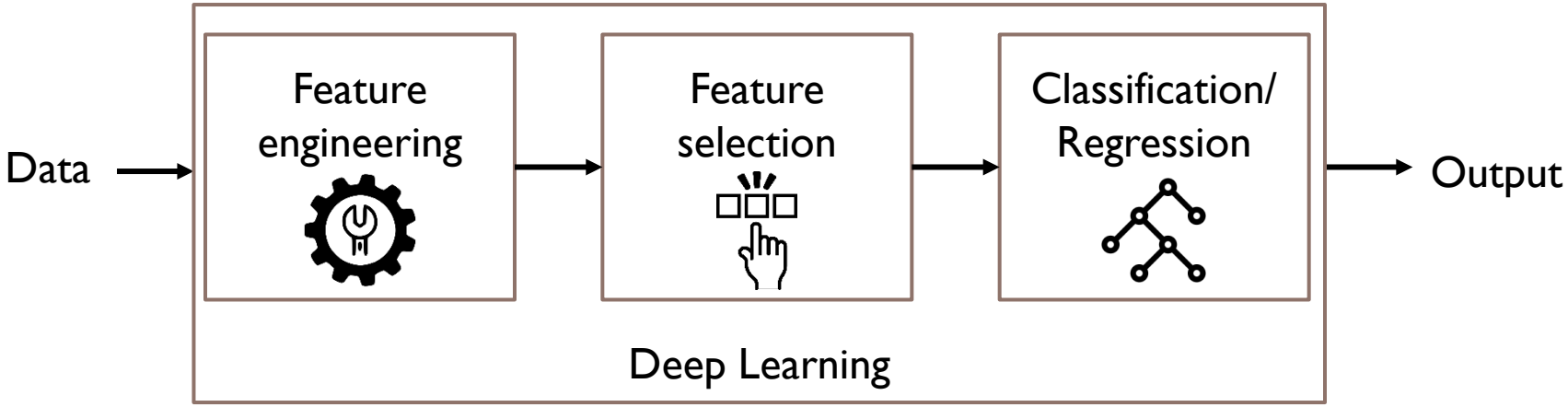
- The classical approach to ML:



- For example: Predicting house prices
  - Data → Webpages from realestate.co.nz
  - Extracted features → house's price, localisation, date of construction, size, owner's name, age, etc…
  - What features are relevant?
  - What regression algorithm?

Making these choices usually requires expert knowledge!

Deep Learning



- No need for manual feature engineering and feature selection.
- Feature generation and classification are both done by the same algorithm (Deep Neural Network), with the data as input.
- Scale well with the amount of training data but requires more computing resources.

Deep Learning and AI systems

- The success of Deep Learning with Deep Neural Networks (DNNs) in the past decades led to the successes of modern AI systems.

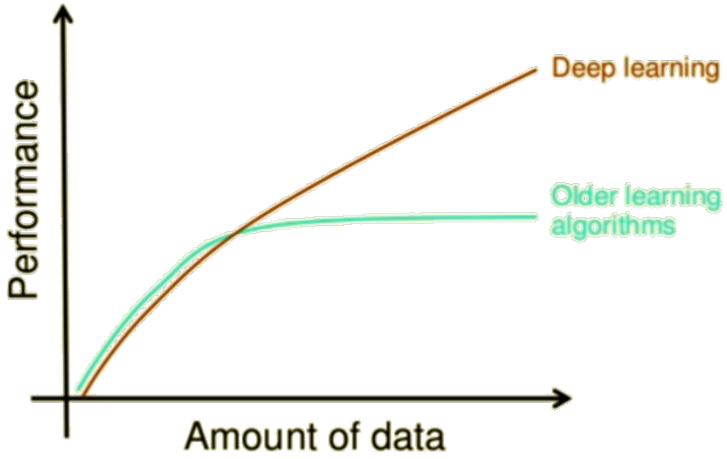
- A lot of the recent major developments in AI systems rely on implementations using Deep Learning and DNNs (e.g., Large Language Models).

- We do not teach much about classical ML algorithms (e.g., random forest, SVM) in COMPSCI 714, but if you want to learn about them, you can take COMPSCI 762.

The death of classical Machine Learning?

The death of classical Machine Learning?

- Classical Machine Learning might still useful in some situations:



Small datasets

Low computational resources

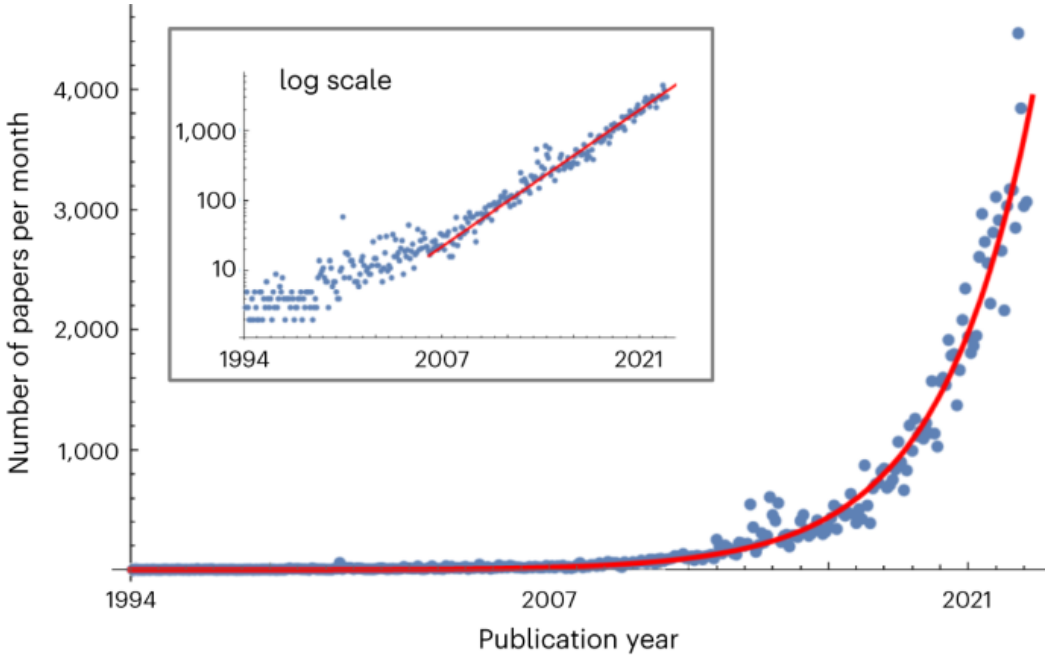
Ease of interpretation

Implementation choices for the course

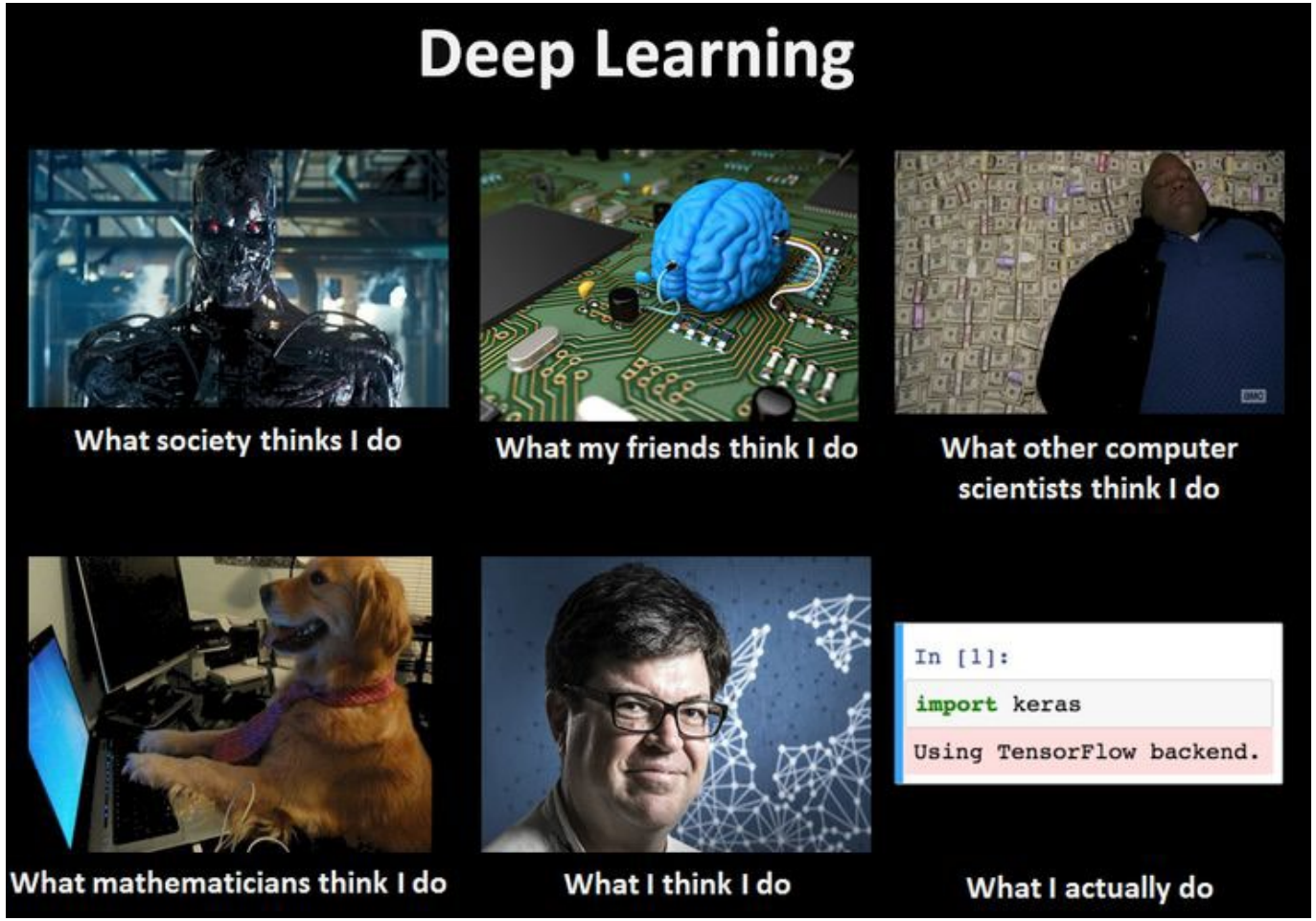
- Two main Python frameworks for Deep Learning implementation:
  - Pytorch (popular in research)
  - TensorFlow (popular in industry)
- There exists a huge amount of DNN open-source architectures.
- We will study a few well-known architectures, but a lot of variants exists.
- Most modern AI solution design is about choosing an existing architecture, relevant to the problem you are trying to solve, and « adapting » it to your problem (fine tuning).

A fast-paced field

- This course will give you a general understanding of the field.
- We will study some historically sucessful architectures, and some of the current State of The Art (SoTA) architectures.
- However, it is crutial that you build habits to keep up-to-date with the rapid developments of the field.



The number of papers published per month in the arXiv categories of AI and ML are growing exponentially.

Krenn, M., Buffoni, L., Coutinho, B. *et al.* Forecasting the future of artificial intelligence with machine learning-based link prediction in an exponentially growing knowledge network. *Nat Mach Intell* **5**, 1326–1335 (2023). https://doi.org/10.1038/s42256-023-00735-0

We hope you enjoy the course!